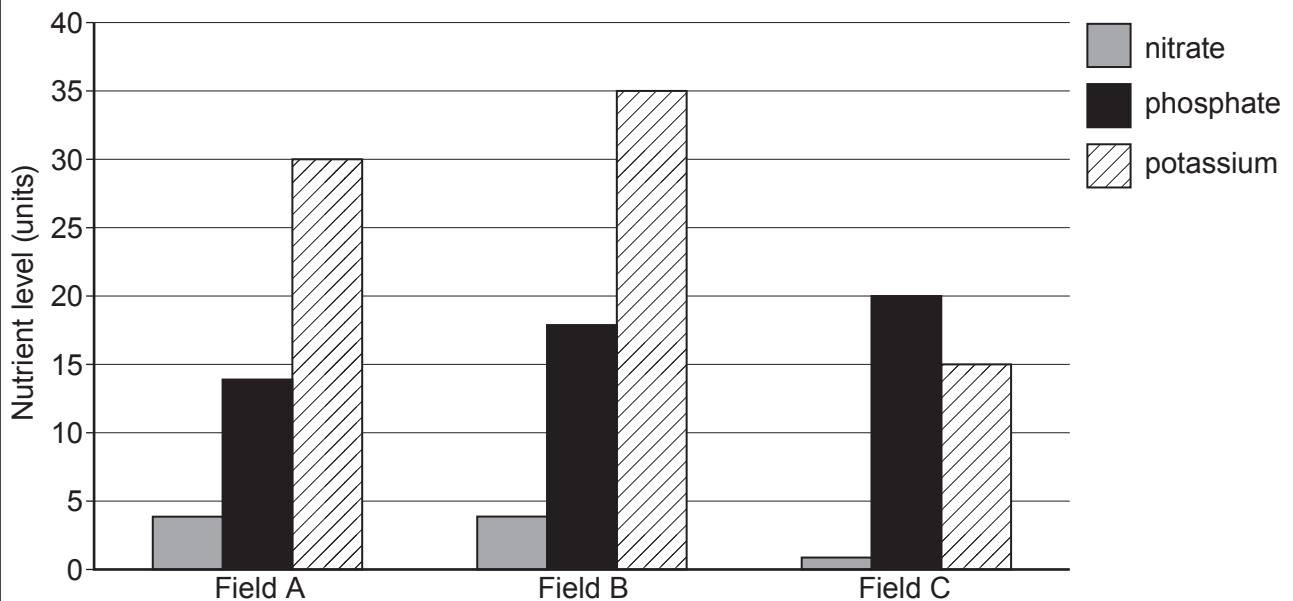


5. A farmer wanted to grow soybean crops on three fields. She analysed soil samples for nitrate, phosphate and potassium. The results are shown below.



Soybeans will show healthy growth if the nutrient levels in the soil are above a threshold value. The plants will show signs of mineral deficiency if the nutrient levels in the soil are below these values.

The nitrate, phosphate and potassium threshold values for soybeans are shown in the table below.

Nutrient	Threshold value (units)
nitrate	2
phosphate	16
potassium	28

- (a) Use the information above to answer the following questions.

- (i) State which field would be the most suitable to grow soybean crops. Give a reason for your answer. [2]

Field:

Reason:

.....



- (ii) State which field would result in soybeans with yellow leaves and stunted growth. Give a reason for your answer. [2]

Field:

Reason:

- (iii) State how the farmer ensures that her soybean plants do not suffer from mineral deficiencies in any field used to grow them. [1]

- (b) Some countries have banned the growing of genetically modified (GM) crops. The table below gives some arguments for and against the use of GM crops. Place a tick (✓) next to **two** arguments **against** growing GM crops. [2]

Argument	Against (✓)
cheaper food	
increased crop yield	
could cross-pollinate with weeds	
long-term effects unknown	

- (c) Selective breeding is used by large companies to increase the yield of soybeans. State **two** disadvantages of using crops developed by a selective breeding process. [2]

1.

2.

